



Dress Rehearsal Problem A

Sum of Three Cubes

Time limit: 2 seconds

In 2019, a team of mathematicians found three cube numbers that sum up to 42 using over a million hours of computing time. With this breakthrough, we have found three cube numbers that sum up to all non-negative integers less than 100 that are not equal to 4 or 5 modulo 9. It has been proven in 1992 that an integer equals to 4 or 5 modulo 9 cannot be represented as a sum of three cube numbers. In other words, for each $0 \leq k < 100$, we have found the triples (x, y, z) such that $x^3 + y^3 + z^3 = k$, or we have proved that no such triplet exists.

The following is a table of one of the values of (x, y, z) that satisfies $x^3 + y^3 + z^3 = k$ for each $0 \leq k \leq 50$ where $k \not\equiv 4 \pmod{9}$ and $k \not\equiv 5 \pmod{9}$.

k	x	y	z
0	0	0	0
1	0	0	1
2	0	1	1
3	1	1	1
6	-1	-1	2
7	0	-1	2
8	0	0	2
9	0	1	2
10	1	1	2
11	-2	-2	3
12	7	10	-11
15	-1	2	2
16	-511	-1609	1626
17	1	2	2
18	-1	-2	3
19	0	-2	3
20	1	-2	3
21	-11	-14	16
24	-2901096694	-15550555555	15584139827
25	-1	-1	3
26	0	-1	3
27	0	0	3
28	0	1	3
29	1	1	3
30	-283059965	-2218888517	2220422932
33	8866128975287528	-8778405442862239	-2736111468807040
34	-1	2	3
35	0	2	3
36	1	2	3
37	0	-3	4
38	1	-3	4
39	117367	134476	-159380
42	-80538738812075974	80435758145817515	12602123297335631
43	2	2	3
44	-5	-7	8
45	2	-3	4
46	-2	3	3



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47	6	7	-8
48	-23	-26	31

Reading a long table is a tedious job, so you would like to create a program that takes an integer n that matches one of the values of k in the table above. Your program should produce three integers x , y , and z . The values of x , y , and z must satisfy $x^3 + y^3 + z^3 = n$, not less than -10^{18} , and not more than 10^{18} . Note that the values of x , y , and z are not required to match the table above.

Input

The first line of input contains one integer n ($0 \leq n \leq 50$, $n \not\equiv 4 \pmod{9}$ and $n \not\equiv 5 \pmod{9}$).

Output

Output three integers x , y , and z that satisfies $x^3 + y^3 + z^3 = n$ and $-10^{18} \leq x, y, z \leq 10^{18}$. If there is more than one solution, you can output any of them. As shown in the above table, it can be shown that at least one solution exists.

Sample Input #1

2

Sample Output #1

1214928 3480205 -3528875

Explanation for the sample input/output #1

Other answers such as $x = 3737830626090$, $y = 1490220318001$, and $z = -3815176160999$ are also accepted.